2015
COMPUTER SCIENCE AND ENGINEERING
Paper – CSE–103
(Advanced Database Management Systems)
Full Marks – 100

The figures in the margin indicate full marks

Candidates are required to give their answers in their own words as far as practicable

Answer ***Question no. 1*** and ***any four*** from the rest

1. Comment critically on the following statements and justify your opinion (***any four***) : 5×4

(a) Join operation should be done after all other operations for distributed database systems.

(b) Query processing for distributed databases involves lower communication cost in general.

(c) Data in data warehouses are usually read-only in nature.

(d) A distributed database is not necessarily a spatial database.

(e) Isolation facilitates concurrent execution of transactions.

(f) Relational data model may not be very efficient for a semi-structured database application.

2. (a) Define min-term predicate.

(b) Briefly describe the methodology for identification of primary horizontal fragments towards the design of a distributed database system.

(c) Explain the significance of derived fragmentation towards efficient design of a distributed database system, using a suitable example.

(d) In the context of distributed database design, why is it important to identify which part of application data are read-only items ? 2+6+8+4

3. (a) Explain the role of semi-join for query optimization in a distributed environment using an example.

(b) Zeus Healthcare Society (ZHS) has a chain of hospitals in big cities across India. The information to be stored is given below:

- Hospital information: hospital id, city, number of beds, bed charges.
- Patient information: patient id, name, address, phone number

[Turn Over]

- Occupancy: patient id, hospital id, bed number, admission date, release date

Design a distributed database solution for hospital bed reservation system to be accessed from multiple terminals in multiple wards located in different hospitals of ZHS considering the following frequent queries/operations:

- Request for reservation of bed. If a patient wants to book multiple beds in the same or more than one hospital of ZHS, for different periods, separate reservation is to be done for each hospital bed and period of booking.
- Availability of bed - the system is to check for availability of bed in a particular hospital for a period.
- Revenue query - the system is to be used to enquire about the revenue generated in a single hospital or in the entire chain for a period.

Your design should include the definition of global schema, fragmentation schema and allocation schema. State your assumptions, if any. 8+12

4. (a) What is distribution transparency ? How is it achieved for distributed databases in multiple levels ?

(b) State two main disadvantages of distributed databases.

(c) Define physical image of a global relation at a site.

(d) Explain the role of local mapping schema towards integration of a heterogeneous multi-site database environment.

(e) Explain with an example how a single updation operation on an single data item could result in multiple updation operations on several sites when the database is physically distributed.

(f) What is semi join ? 4+3+1+6+5+1

5. (a) Define the following terms in the context of building a data warehouse :

(i) Extraction (ii) Transformation (iii) Loading (iv) Integration

(b) Why granularity of data is an important aspect towards efficient designing of a data warehouse ?

(c) How Snow-Flake is different from a Star Schema for data warehouse?

(d) A cube model has data in four dimensions for a particular subject orientation. The cardinalities for the four dimensions are 3, 4, 5 and 4 respectively.

(i) How many Fact and Dimension Tables are required in an equivalent Star schema ?

(ii) State and explain the minimum number of tuples that would occur in each of Dimension and Fact Tables in your design.

(e) How do you relate a lattice with a data warehouse ? 4+1+3+6+6

6. (a) What is semi-structured data ?

(b) Why newer conceptual data models are required for semi-structured data ?

(c) ITRA-funded remote healthcare project CUDFIKJ requires to store data pertaining to different patients, their ailment, doctors and other care givers. A patient may need advices of one or more doctor(s), depending on the disease and availability of resources. The discomforts or symptoms recorded for patients vary a lot from case to case.

We assume that a Hyper file named *PatientComplaints* stores the basic information like patient name, address, age followed by one or more discomforts. The parameters for discomfort change on its nature. As for example, if a patient complains about pain, information is recorded on the organ where the pain is felt, nature of pain (continual or periodic, etc.), etc. On the other hand, if the patient complains about fever, it's important to store the body temperature.

The employee profile too varies depending on the types of employees and their job functionality. We assume that a Hyper file named *HealthServices* stores the basic information like service-title, doctor-id, major-Symptom, and cost for different services. Assume that some of the entries in the Hyper file *HealthServices* point at some other entries as FRT (frequently required together), if such services are often requested together.

(i) Write a query using Hyper file model to list the name and age for all the Patients treated in the CUDFIKJ project.

(ii) Write a query using Hyper file model to extract FRT services for patients coming with chest pain complaints as major symptom. Your query must extract all the items that are linked up to third level considering chest-pain in level 1.

(iii) State your assumptions clearly for the additional Hyper files in the database other than the two mentioned above that you may need to define for answering the queries. 2+4+[4+6+4]

7. (a) How primary-copy locking is useful for resolving conflicts in a distributed database?

(b) Explain the roles of distributed transaction manager (DTM) and local transaction manager (LTM) towards transaction management.

(c) A "commit" operation in database systems ensures that the changes in the database are permanent. In the 2-phase commit protocol for transaction management, two different types of "commit" operations are performed at the coordinating site and at the participating sites. The coordinating

site may decide to abort a transaction even if one or more participating site is ready for "local commit". Explain whether this is consistent with the ACID properties for transaction management.

(d) "Isolation facilitates concurrent execution of transactions" - justify the correctness of the statement.

(e) How the atomicity of a transaction is ensured under 2-phase commit protocol when one of the sites fails after the coordinator issues a global commit message ?

(f) Compare primary copy locking and distributed 2-phase locking.

2+4+3+5+3+3